

OS Practical 7

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;

void function()
{
}

int main()
{
    static int static_var = 20;
    int stack_var = 30;
    int *heap_var = malloc(sizeof(int));

    *heap_var = 40;

    printf("Process ID : %d\n", getpid());

    printf("Code address   : %p\n", (void *)function);
    printf("Global address  : %p\n", (void *)&global_var);
    printf("Static address   : %p\n", (void *)&static_var);
    printf("Heap address     : %p\n", (void *)heap_var);
    printf("Stack address    : %p\n", (void *)&stack_var);

    printf("\nPress Enter to exit...\n");
    getchar();

    free(heap_var);
    return 0;
}
```

Output:

```
nishanth@LAPTOP-6Q3ON07H:~/practical7$ gcc program1.c -o program1
nishanth@LAPTOP-6Q3ON07H:~/practical7$ ./program1
Process ID : 706
Code address   : 0x63c08e117209
Global address  : 0x63c08e11a010
Static address  : 0x63c08e11a014
Heap address    : 0x63c0cc550010
Stack address   : 0x7ffd61d818bc

Press Enter to exit...
```

```

#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
    FILE *fp;
    char filename[100];
    char line[500];

    if (argc != 2)
    {
        printf("Usage: %s <PID>\n", argv[0]);
        return 1;
    }

    sprintf(filename, "/proc/%s/maps", argv[1]);

    fp = fopen(filename, "r");

    if (fp == NULL)
    {
        printf("Unable to open %s\n", filename);
        return 1;
    }

    printf("Memory Map of Process %s\n", argv[1]);
    printf("-----\n");

    while (fgets(line, sizeof(line), fp) != NULL)
    {
        printf("%s", line);
    }

    fclose(fp);

    return 0;
}

```

Output:

Memory Map of Process 799

```

-----
62a22d983000-62a22d984000 r--p 00000000 08:30 13625 /home/nishanth/practical7/program1
62a22d984000-62a22d985000 r-xp 00001000 08:30 13625 /home/nishanth/practical7/program1
62a22d985000-62a22d986000 r--p 00002000 08:30 13625 /home/nishanth/practical7/program1
62a22d986000-62a22d987000 r--p 00002000 08:30 13625 /home/nishanth/practical7/program1
62a22d987000-62a22d988000 rw-p 00003000 08:30 13625 /home/nishanth/practical7/program1
62a24f28f000-62a24f2b0000 rw-p 00000000 00:00 0 [heap]
76cad9c00000-76cad9c28000 r--p 00000000 08:30 35697 /usr/lib/x86_64-linux-gnu/libc.so.6
76cad9c28000-76cad9dc0000 r-xp 00028000 08:30 35697 /usr/lib/x86_64-linux-gnu/libc.so.6
76cad9dc0000-76cad9e0e000 r--p 001c0000 08:30 35697 /usr/lib/x86_64-linux-gnu/libc.so.6
76cad9e0e000-76cad9e12000 r--p 0020e000 08:30 35697 /usr/lib/x86_64-linux-gnu/libc.so.6
76cad9e12000-76cad9e14000 rw-p 00212000 08:30 35697 /usr/lib/x86_64-linux-gnu/libc.so.6
76cad9e14000-76cad9e21000 rw-p 00000000 00:00 0
76cad9e61000-76cad9e64000 rw-p 00000000 00:00 0
76cad9e6c000-76cad9e6e000 rw-p 00000000 00:00 0
76cad9e6e000-76cad9e72000 r--p 00000000 00:00 0 [vvar]
76cad9e72000-76cad9e74000 r--p 00000000 00:00 0 [vvar_vclock]
76cad9e74000-76cad9e76000 r-xp 00000000 00:00 0 [vdso]
76cad9e76000-76cad9e77000 r--p 00000000 08:30 35694 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
76cad9e77000-76cad9ea6000 r-xp 00001000 08:30 35694 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
76cad9ea6000-76cad9eb1000 r--p 00030000 08:30 35694 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
76cad9eb1000-76cad9eb3000 r--p 0003b000 08:30 35694 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
76cad9eb3000-76cad9eb4000 rw-p 0003d000 08:30 35694 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
76cad9eb4000-76cad9eb5000 rw-p 00000000 00:00 0
7fff5cb76000-7fff5cb97000 rw-p 00000000 00:00 0 [stack]

```